

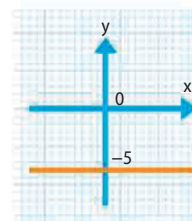
Using graphs to solve quadratic equations

A quadratic equation can be solved by drawing a graph. If a quadratic equation has a y value that is not 0, it can be solved by drawing both a quadratic and a linear graph (the linear graph is of the y value that is not 0) and finding where the two graphs cross. The solutions to the equation are the x values where the two graphs cross.

This equation has two parts: a quadratic equation on the left and a linear equation on the right. To find the solutions to this equation, draw the quadratic and linear graphs on the same axes. To draw the graphs, it is necessary to find sets of x and y values for both sides of the equation.

$$\underbrace{-x^2 - 2x + 3}_{\substack{\text{quadratic part of equation} \\ \text{y values for quadratic} \\ \text{part of equation are} \\ \text{dependent on value} \\ \text{of } x}} = \underbrace{-5}_{\substack{\text{linear part of equation} \\ \text{y values for linear} \\ \text{part of equation} \\ \text{are all } -5}}$$

$$y = -x^2 - 2x + 3$$



$\triangle y = -5$
This graph is very simple: whatever value x takes, y is always -5 . This means that the graph is a straight horizontal line that passes through the y axis at -5 .

Find values of x and y for the quadratic part of the equation using a table. Choose x values either side of 0 and split the equation into parts ($-x^2$, $-2x$, and $+3$). Work out the value of each part for each value of x , then add the values of all three parts to find the y value for each x value.

choose some values of x around 0

$y = -x^2 - 2x + 3$, so it is difficult to work out the y values right away

work out x^2 first then put a minus sign in front to give values

work out $-2x$ in this column

$+3$ is the same for each x value

x	$-x^2$	$-2x$	3	y
-4	-16	+8	+3	
-3	-9	+6	+3	
-2	-4	+4	+3	
-1	-1	+2	+3	
0	0	0	+3	
1	-1	-2	+3	
2	-4	-4	+3	

add values in each purple row to find y values

x	$-x^2$	$-2x$	3	y
-4	-16	+8	+3	-5
-3	-9	+6	+3	0
-2	-4	+4	+3	3
-1	-1	+2	+3	4
0	0	0	+3	3
1	-1	-2	+3	0
2	-4	-4	+3	-5

y is the sum of numbers in each purple row

+

+

=

$\triangle$ Values of x

Each value of y depends on the value of x . Choose a number of values for x , and work out the corresponding values of y . It is easiest to include 0 and values of x that are on either side of 0.

$\triangle$ Different parts of the equation

The equation has 3 different parts: $-x^2$, $-2x$, and $+3$. Work out the values of each part of the equation for each value of x , being careful to pay attention to whether the values are positive or negative. The last part of the equation, $+3$, is the same for each x value.

$\triangle$ Corresponding values of y

Finally, add the three parts of the equation together to find the corresponding values of y for each x value. Make sure to pay attention to whether the different parts of the equation are positive or negative.